

Engineering Mechanics MEL 1010
Major Exam

Total Time: 3 hrs

Total Marks: 140

Total Number of Questions: 12

If you complete the major exam early, submit your answer sheet and leave the examination room quietly.

1. The slotted arm OA forces the small pin to move in the fixed spiral guide defined by $r = K\theta$. Arm OA starts from rest at $\theta = \pi/4$ and has a constant counterclockwise angular acceleration $\ddot{\theta} = \alpha$. Determine the magnitude of the acceleration of the pin P when $\theta = 3\pi/4$.

Hint: Use polar coordinates and note that it is **constant** angular acceleration.

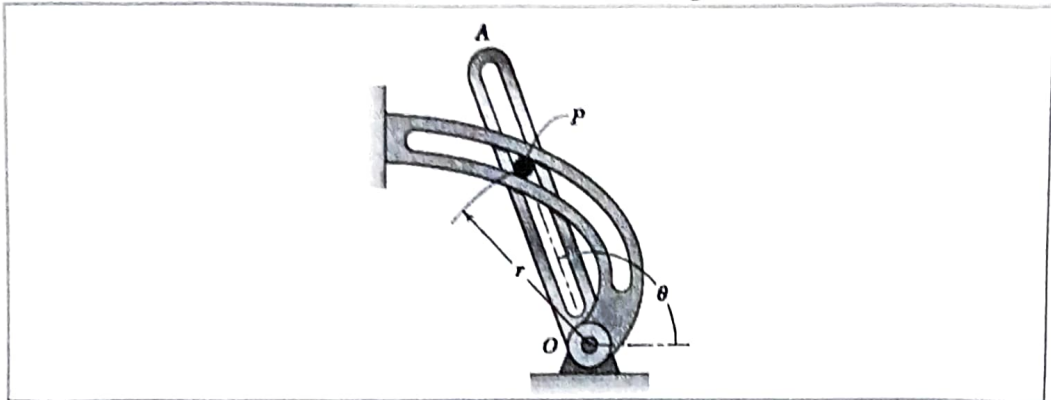


Fig. 1

Marks:10

2. A car is climbing the hill of slope θ_1 at a constant speed v . If the slope decreases abruptly to θ_2 at point A , determine the acceleration a of the car just after passing point A if the driver does not change the throttle setting or shift into a different gear.

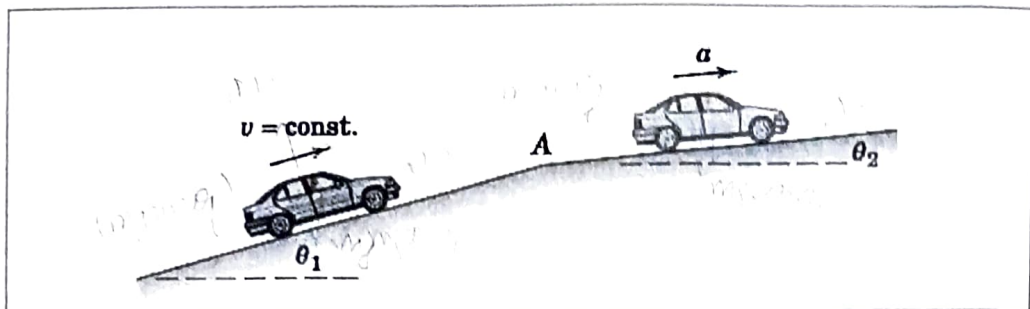


Fig. 2

Marks:5

3. A particle of mass m is released from rest in position A and then slides down the smooth vertical-plane track. Determine its angular momentum about both points A and D (a) as it passes position B and (b) as it passes position C .

Hint: Angular momentum for a single particle is moment of linear momentum.

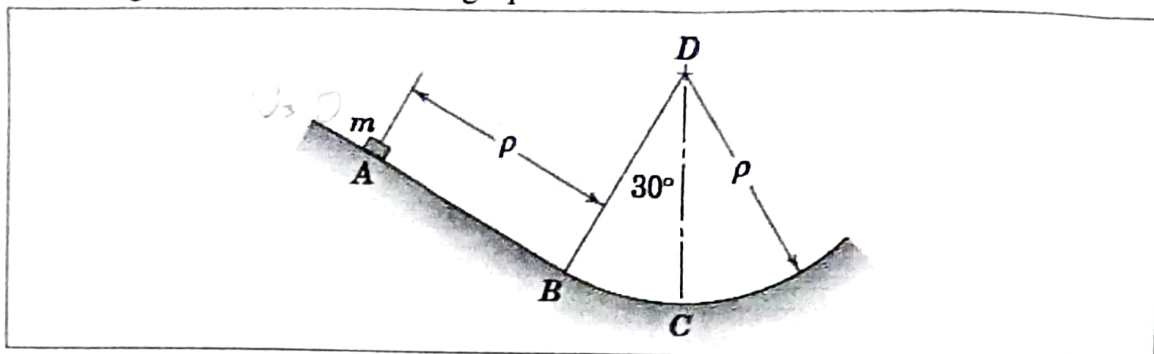


Fig. 3

Marks: 5+5=10

4. At the instant when $x_A = 0.85L$, the velocity of the slider at A is $v = 2$ m/s to the right. Determine the corresponding velocity of slider B and the angular velocity ω of bar AB if $L = 0.8$ m.

Hint: You may find instantaneous center of zero velocity.

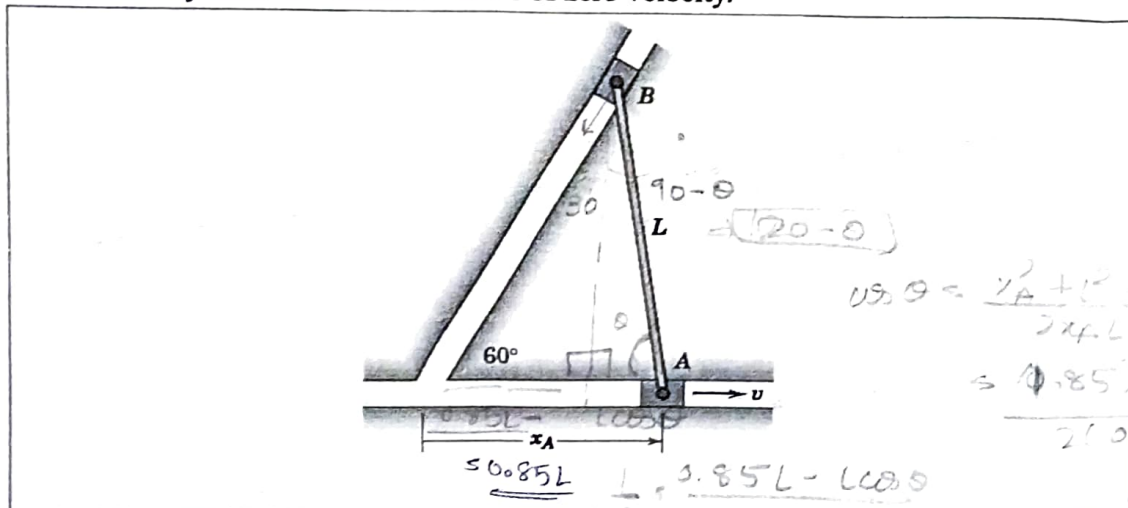


Fig. 4

Marks: 10

$$\frac{1}{2} = \frac{0}{1.84} \\ 0.92$$

+ 35

35. The circular disk rolls to the left without slipping. If $a_{A/B} = -2.7 \text{ j m/s}^2$, determine the velocity and acceleration of the center O of the disk.
Hint: A and B are on one rigid body. Relative motion between two points on a rigid body can be considered as a pure rotation about one point.

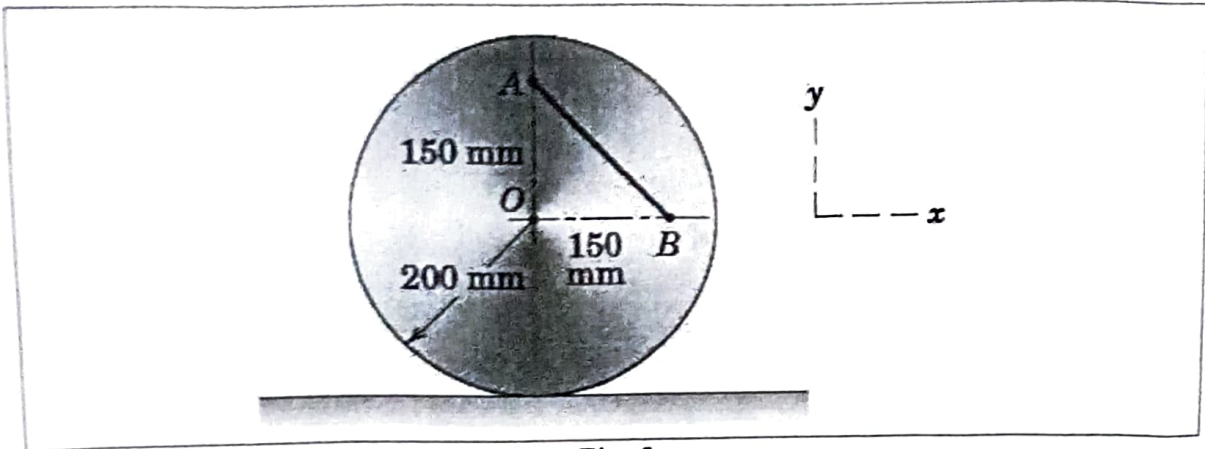


Fig. 5

Marks: 10

6. The disk rotates about a fixed axis through O with angular velocity $\omega = 5 \text{ rad/sec}$ and angular acceleration $\alpha = 3 \text{ rad/sec}^2$ at the instant represented, in the directions shown. The slider A moves in the straight slot. Determine the absolute velocity and acceleration of A for the same instant, when $y = 8 \text{ cm}$, $\dot{y} = -24 \text{ cm/sec}$, and $\ddot{y} = 30 \text{ cm/sec}^2$.
Hint: Use fixed frame of reference $X-Y$ and rotating frame of reference $x-y$. x -coordinate of A is not changing in rotating frame of reference and only y -coordinate changes.

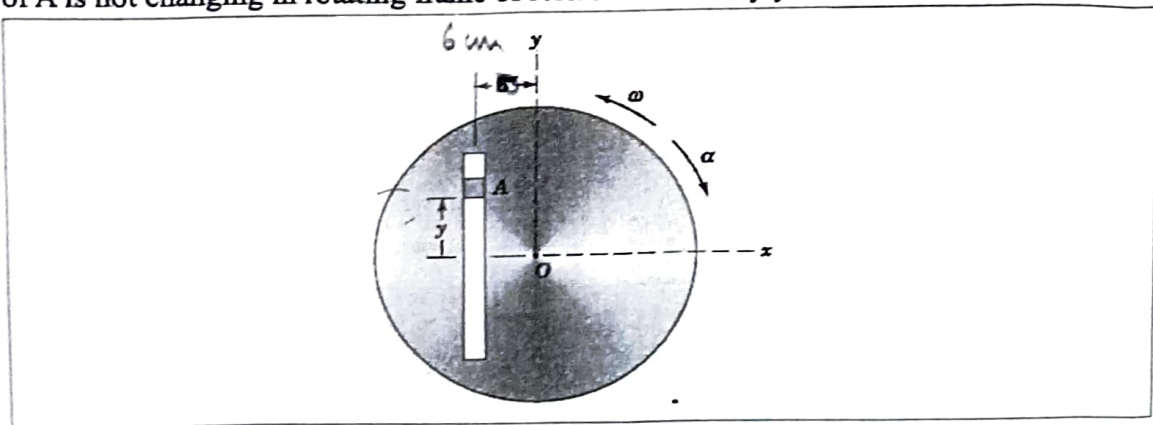


Fig. 6

Marks: 10

$$\underline{a_A = a_O + \alpha \times r + \omega \times (\omega \times r)}$$

$$\begin{aligned} V_A &= V_O + \omega \times r + V_{rel} \\ a_A &= a_O + \dot{\omega} \times r + \omega \times (\omega \times r) \\ &\quad + 2\omega \times V_{rel} + a_{rel} \end{aligned}$$

7. The disk rotates about a fixed axis through O with angular velocity $\omega = 5 \text{ rad/sec}$ and angular acceleration $\alpha = 3 \text{ rad/sec}^2$ in the directions shown at a certain instant. The small sphere A moves in the circular slot, and at the same instant, $\beta = 30^\circ$, $\dot{\beta} = 2 \text{ rad/sec}$, and $\ddot{\beta} = -4 \text{ rad/sec}^2$. Determine the absolute velocity and acceleration of A at this instant.

Hint: Use rotating frame of reference $x-y$.

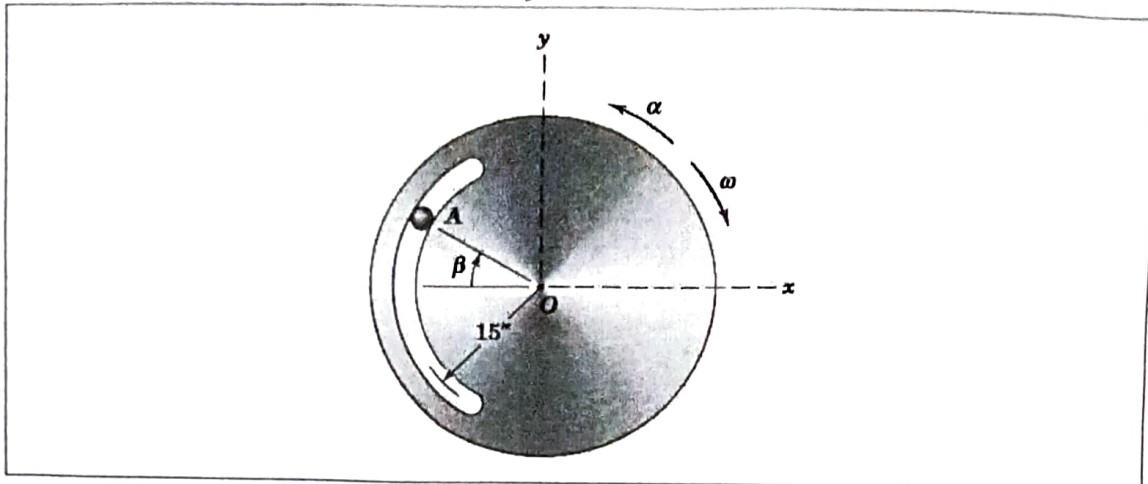


Fig. 7

Marks:10

8. The uniform 200-kg bar AB is raised in the vertical plane by the application of a constant couple $M = 3 \text{ kN.m}$ applied to the link at C . The mass of the links is small and may be neglected. If the bar starts from rest at $\theta = 0$, determine the magnitude of the force supported by the pin at A as the position $\theta = 60^\circ$ is passed.

Hint: Think whether this is curvilinear translation or rotation of the body about fixed axis or general plane motion. Apply corresponding force and moment equation about center of mass and about any other point.

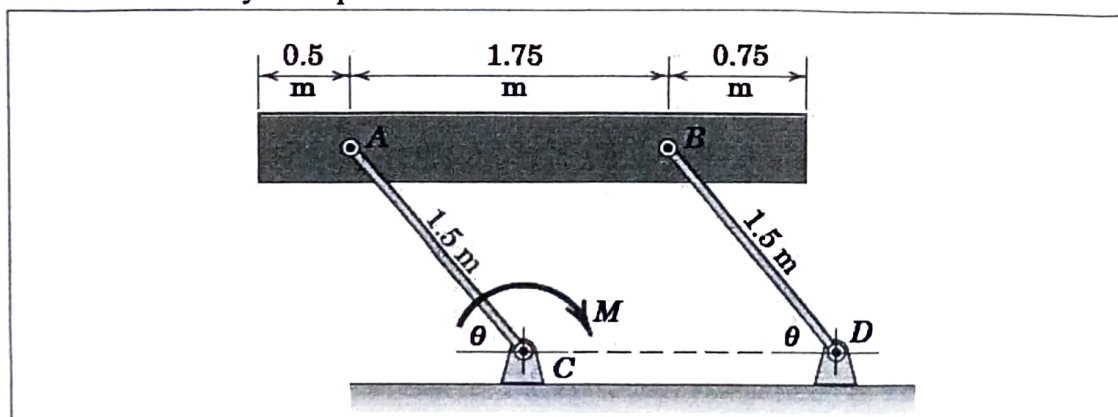


Fig. 8

Marks:15

9. The wheel consists of a 4-kg rim of 250-mm radius with hub and spokes of negligible mass. The wheel is mounted on the 3-kg yoke OA with mass center at G and with a radius of gyration about O of 350 mm. If the assembly is released from rest in the horizontal position shown and if the wheel rolls on the circular surface without slipping, compute the velocity of point A when it reaches at A' .

Hint: Work-Energy Principle

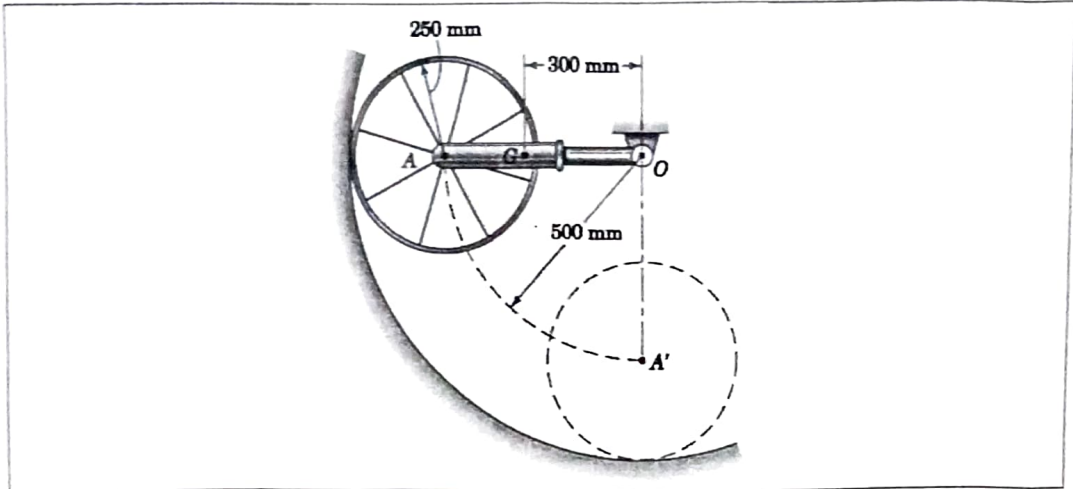


Fig. 9

Marks:15

10. The spacecraft shown has a mass m with mass center G . Its radius of gyration about its z -axis of rotational symmetry is k and that about either the x - or y -axis is k' . In space, the spacecraft spins within its x - y - z reference frame at the rate $p = \dot{\phi}$. Simultaneously, point C on the z -axis moves in a circle about the z_0 -axis with a frequency f (rotations per unit time). The z_0 -axis has a constant direction in space. Determine the angular momentum $\mathbf{H}_G$ of the spacecraft relative to the axes designated. Note that the x -axis always lies in the z - z_0 plane and that the y -axis is therefore normal to z_0 .

Hint: Angular momentum can be expressed easily about principal axes of inertia. Spin velocity is p . $\mathbf{H} = I_{xx}\omega_x + I_{yy}\omega_y + I_{zz}\omega_z$

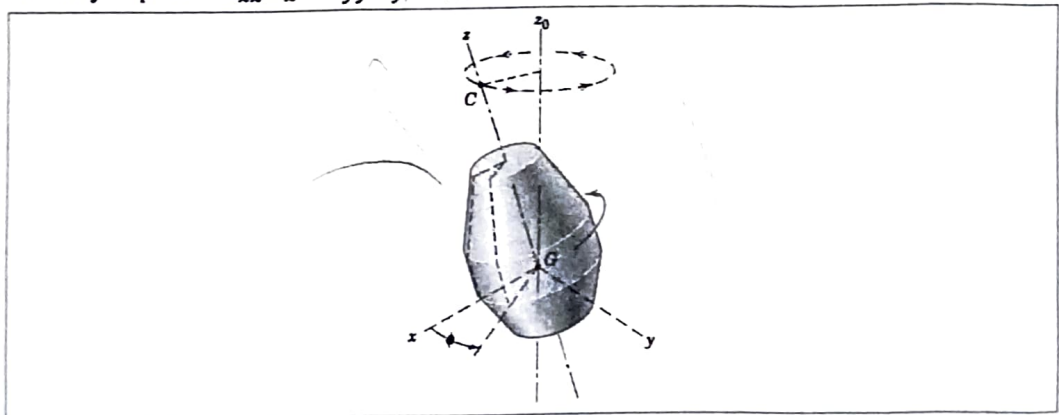


Fig. 10

Marks:15

11. The 8 kg rotor with radius of gyration of 3 cm rotates on ball bearings at a speed of 3000 rev/min about its shaft OG . The shaft is free to pivot about the X -axis, as well as to rotate about the Z -axis. Calculate the vector Ω for precession about the Z -axis. Neglect the mass of shaft OG and compute the gyroscopic couple M exerted by the shaft on the rotor at G .

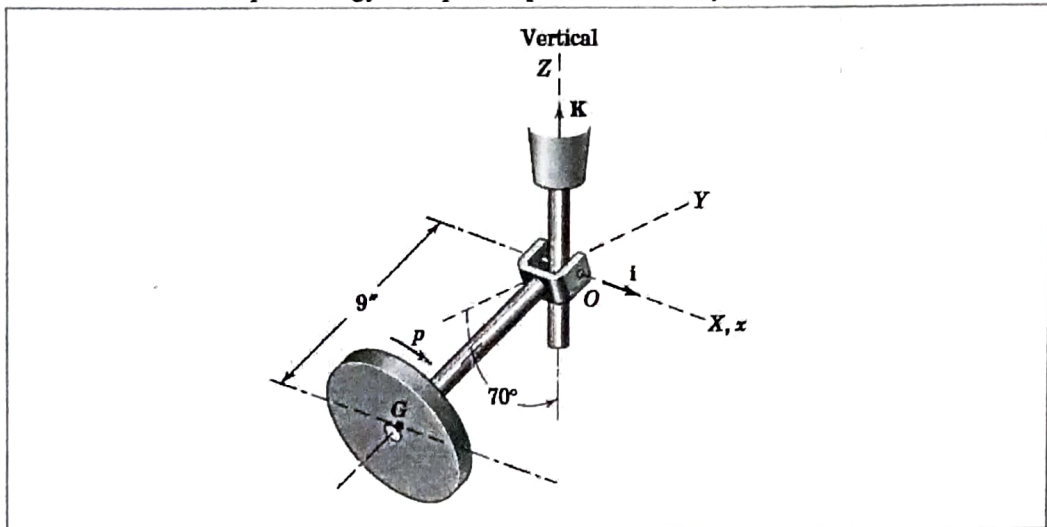


Fig. 11

Marks: 15

12. If the motor pivots about the x -axis at the constant rate $\dot{\gamma} = 3\pi$ rad/sec with no rotation about the Z -axis ($N=0$), determine the angular acceleration α of the rotor and disk as the position $\gamma = 30^\circ$ is passed. The constant speed of the motor is 120 rev/min. Also find the velocity and acceleration if point A , which is on the top of the disk for this position. $OC=10$ cm and $CA=5$ cm.

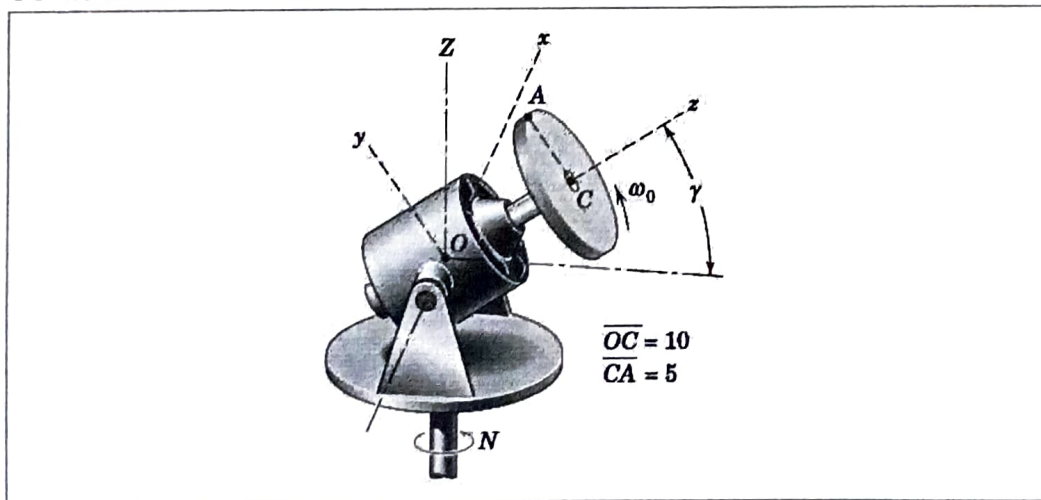


Fig. 12

Marks: 15

After the exam: How to proceed in Q. 12 if N (rotation about Z -axis) has some constant value (not zero)?

BEST WISHES and ENJOY!!

Major Exam: Engineering Mechanics MEL1010

Total Questions: 9

Total Marks: 90 (10*9)

Total time: 3 hours (9AM-12 AM)

Bring your own scientific calculator. If you complete quiz early, submit your answer sheet and leave the room quietly. If you are found using any unfair means, you will be asked to submit the answer sheet and **you should leave the room immediately. Do not argue.** Your answer sheet will not be evaluated.

1. The piston of the hydraulic cylinder gives pin A a constant velocity $v = 3 \text{ ft/sec}^2$ in the direction shown for an interval of its motion. For the instant when $\theta = 60^\circ$, determine $\dot{r}$, $\ddot{r}$, $\dot{\theta}$, and $\ddot{\theta}$ where $r = \overline{OA}$.

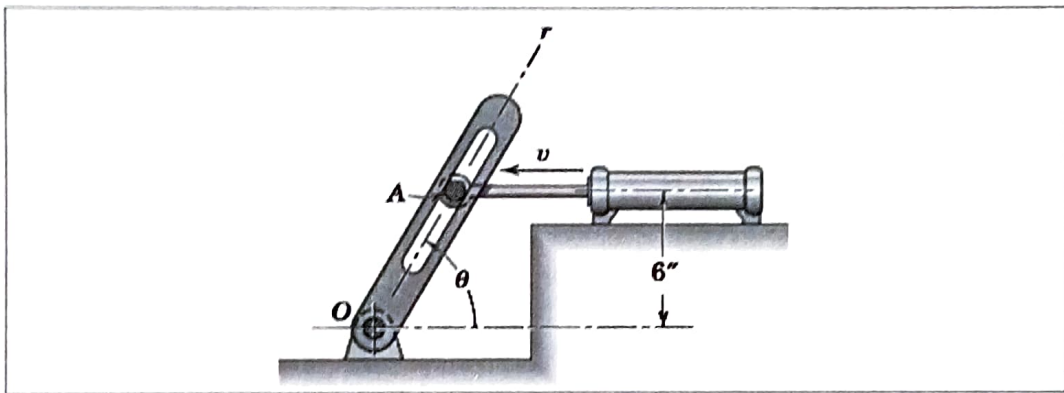


Fig. 1

2. Determine the coefficient of restitution e which will allow the ball to bounce down the steps as shown. The tread and riser dimensions, d and h , respectively, are the same for every step, and the ball bounces the same distance h' above each step. What horizontal velocity v_x is required so that the ball lands in the center of each tread?

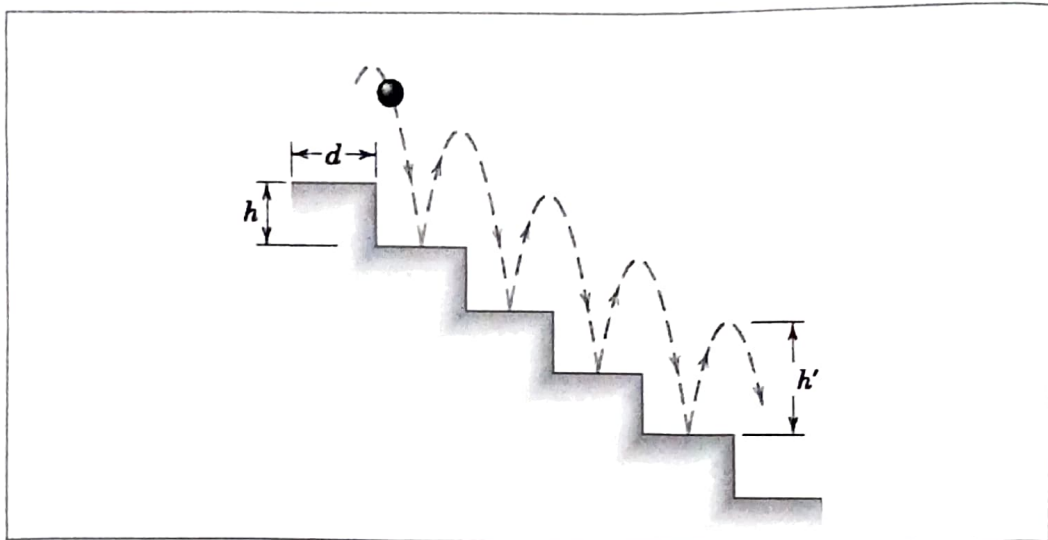


Fig. 2

3. The three small spheres, each of mass m , are secured to the light rods to form a rigid unit supported in the vertical plane by the smooth circular surface. The force of constant magnitude P is applied perpendicular to one rod at its midpoint. If the unit starts from rest at $\theta = 0^\circ$, determine (a) the minimum force $P_{\min}$ which will bring the unit to rest at $\theta = 60^\circ$ and (b) the common velocity v of spheres 1 and 2 when $\theta = 60^\circ$ if $P = 2P_{\min}$.

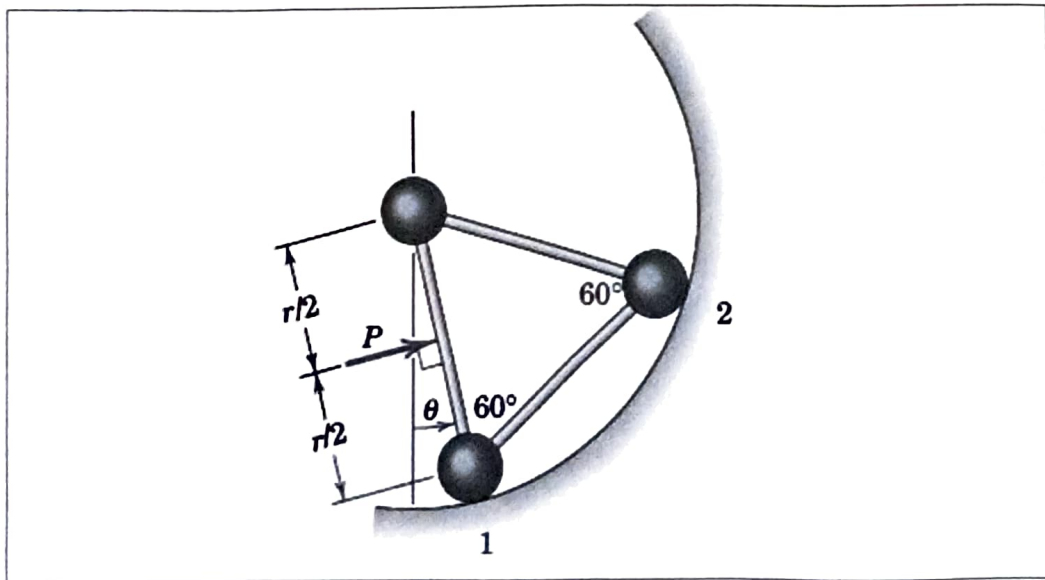


Fig. 3

4. Show that the expressions $v = r\omega$ and $a_t = r\alpha$ hold for the motion of the center O of the wheel which rolls on the concave or convex circular arc, where ω and α are the absolute angular velocity and acceleration, respectively, of the wheel. (*Hint: Allow the wheel to roll a small distance. Be very careful to identify the correct absolute angle through which the wheel turns in each case in determining its angular velocity and angular acceleration.*)

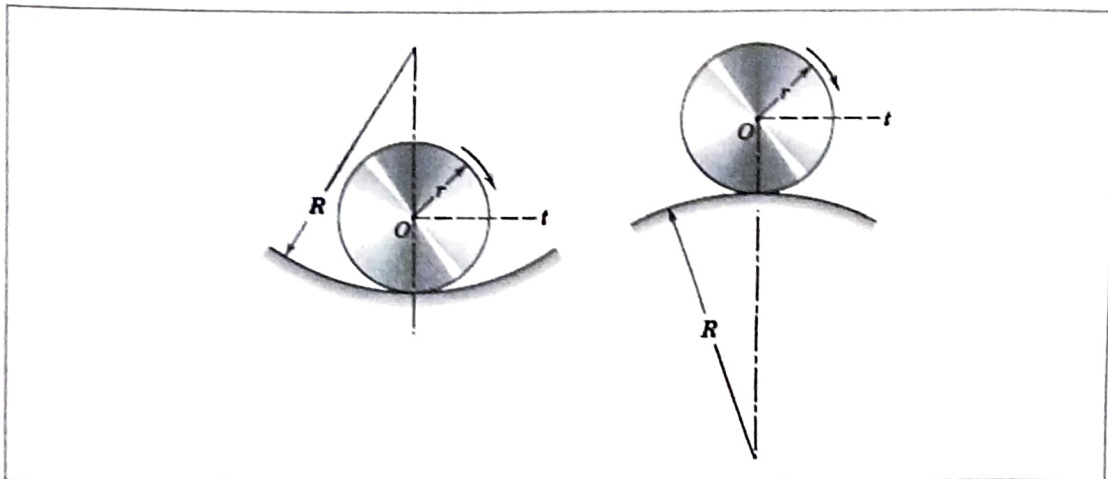


Fig. 4

5. The disk rolls without slipping on the horizontal surface, and at the instant represented, the center O has the velocity and acceleration shown in the figure. For this instant, the particle A has the indicated speed u and time rate of change of speed $\dot{u}$, both relative to the disk. Determine the absolute velocity and acceleration of particle A .

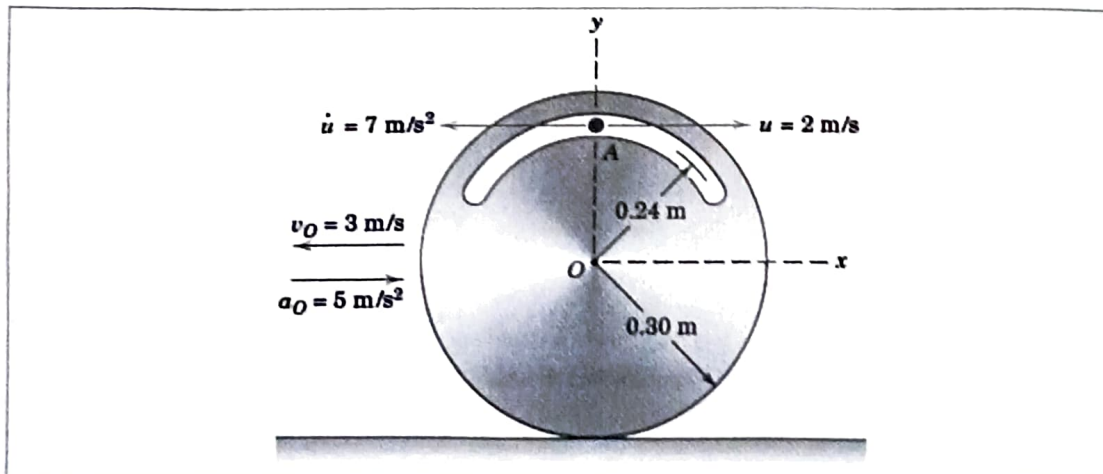


Fig. 5

6. The 25-kg bar BD is attached to the two light links AB and CD and moves in the vertical plane. The lower link is subjected to a clockwise torque $M = 100 \text{ N}\cdot\text{m}$ applied through its shaft at A. If each link has an angular velocity $\omega = 5 \text{ rad/s}$ as it passes the horizontal position, calculate the force which the upper link exerts on the bar at D at this instant. Also find the angular acceleration of the links at this position.

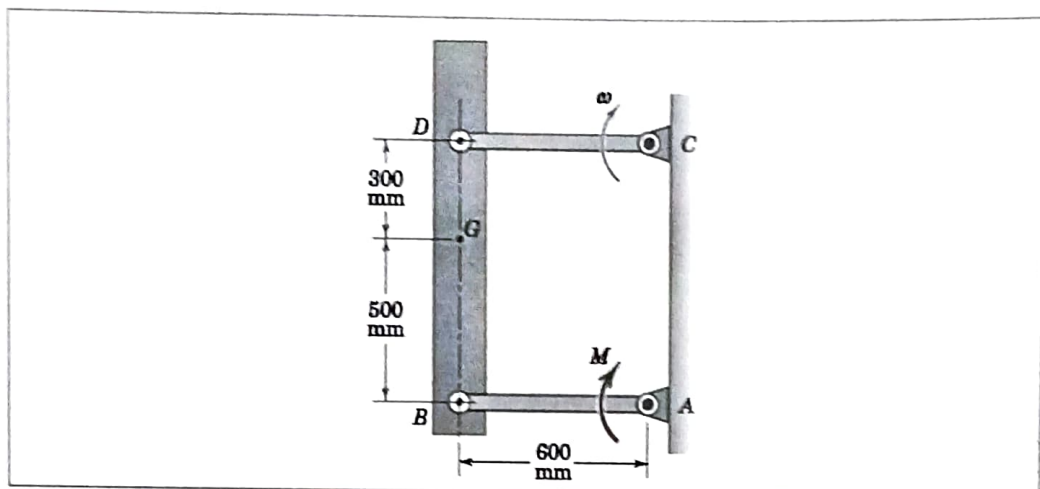


Fig. 6

7. The 60-ft telephone pole of essentially uniform diameter is being hoisted into the vertical position by two cables attached at B as shown. The end O rests on a fixed support and cannot slip. When the pole is nearly vertical, the fitting at B suddenly breaks, releasing both cables. When the angle θ reaches 10° , the speed of the upper end A of the pole is 4.5 ft/sec. From this point, calculate the time which the workman would have to get out of the way before the pole hits the ground. With what speed v_A does end A hit the ground?

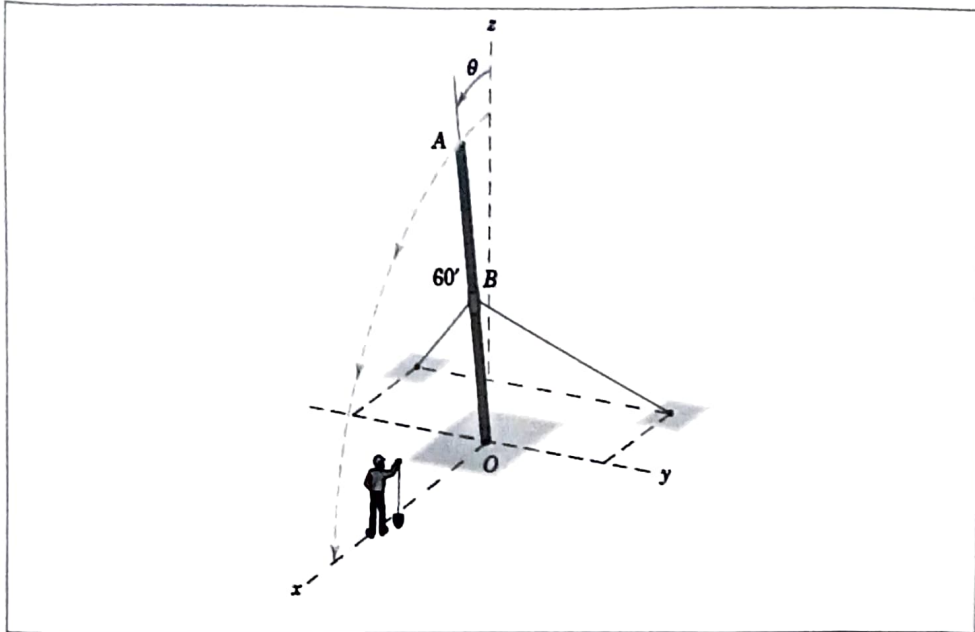


Fig. 7

8. The uniform circular disk of 4-inch radius and small thickness weighs 8 lb and is spinning about its y' -axis at the rate $N = 300 \text{ rev/min}$ with its plane of rotation tilted at a constant angle $\beta = 20^\circ$ from the vertical x-z plane. Simultaneously, the assembly rotates about the fixed z-axis at the rate $p = 60 \text{ rev/min}$. Calculate the angular momentum H_O of the disk alone about the origin O of the x-y-z coordinates. Also calculate the kinetic energy T of the disk.

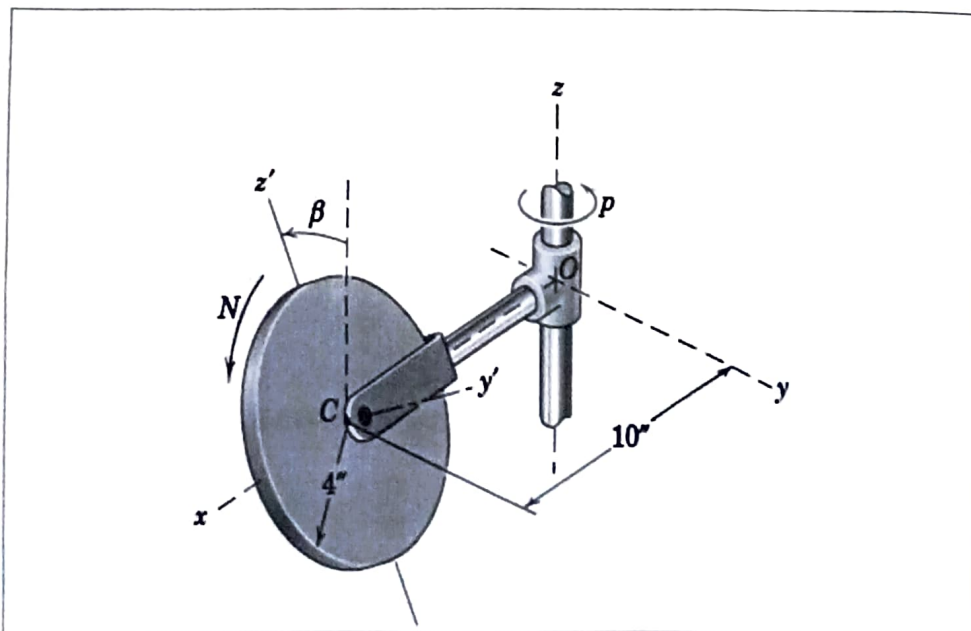


Fig. 8

9. The motor turns the disk at the constant speed $p = 30$ rad/sec. The motor is also unwinding about the horizontal axis B-O (y-axis) at the constant speed $\dot{\theta} = 2$ rad/sec. Simultaneously, the entire assembly is rotating about the vertical axis C-C at the constant rate $q = 8$ rad/sec. For the instant when $\theta = 30^\circ$, determine the angular acceleration α of the disk and the acceleration $\mathbf{a}$ of point A at the bottom of the disk. Axes x - y - z are attached to the motor housing, and plane O- x_1 - y is horizontal.

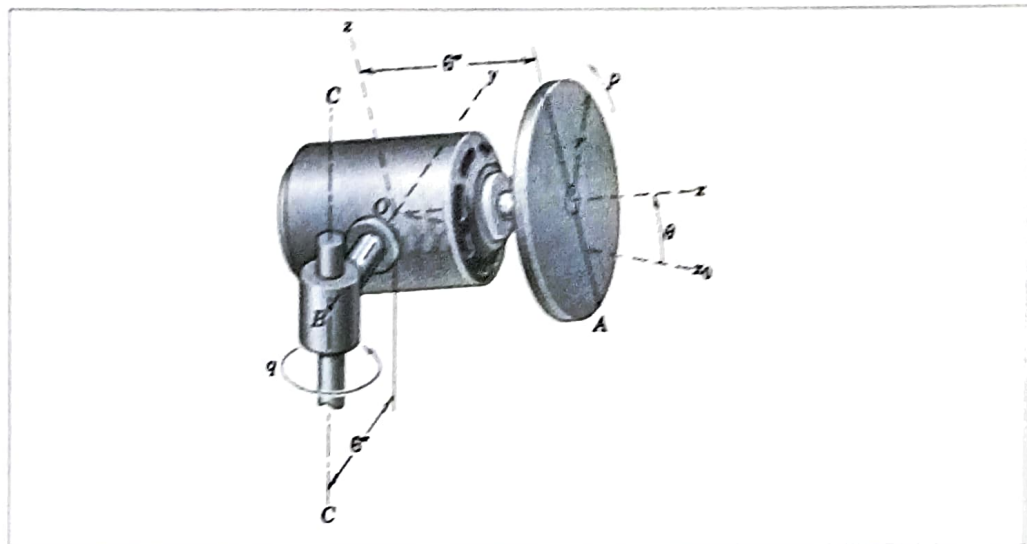


Fig. 9